Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 1–11

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Secure Goods Storage and Anti-Theft Approach using Ethereum Blockchain

Likhitha Amasala[a]*, Mahesh Ponnuru[b], Srideviponmalar P[c]

[a]*School of Computer Science and Engineering,Vellore Institute of Technology, Amaravathi-522237,India*
[b]*School of Computing,SRM Institute of Science and Technology,Potheri-603203,India.*
[c]*Department of Computational Intelligence,School of Computing,Faculty of Engineering and Technology,SRM Institute of Science and Technology,Potheri-603203,India*

Abstract

At present, technological systems lack a secure and transparent method for tracking goods and preventing theft in e-commerce, leading to trust issues and data vulnerabilities. There is a pressing need for a comprehensive solution that integrates Ethereum blockchain, IPFS, and advanced cryptographic techniques to address these challenges and enhance the security and transparency of transactions. This research paper presents a robust system that harnesses the Ethereum blockchain, IPFS (Interplanetary File System), and advanced cryptographic algorithms to create a secure, decentralized approach for tracking goods and preventing theft incidents. Unique identifiers and related information will be sent to the mail of the customer and same should entered by the customer for successful transaction. By assigning unique identifiers to purchased products and employing cryptographic techniques to encrypt sensitive data, our system ensures both user privacy and the creation of an immutable transaction ledger. Users can efficiently manage their purchased goods, block stolen items, and communicate with sellers through an intuitive interface. Additionally, the system provides sellers with a comprehensive transaction history, enhancing accountability and transparency within the supply chain. Through this research, we demonstrate the effectiveness of our blockchain based anti-theft measures, underpinned by Ethereum, IPFS, and cutting-edge cryptographic algorithms, in fostering secure, trustless transactions. This work highlights the transformative potential of blockchain technology and decentralized protocols in revolutionizing security and transparency across diverse sectors.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Ethereum Blockchain; Secure Goods Storage; Anti-theft; Interplanetary File System; Smart Contracts; Ownership verification.

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.190

1. Introduction

In today's age of digitalization and global connectivity, ensuring the security and transparency of goods transactions has become of paramount importance. Traditional methods of tracking transactions often struggle to address evolving challenges, including theft and a lack of transparency. In response to these challenges, blockchain technology has emerged as a transformative solution, offering a decentralized and tamper-resistant platform that has the potential to redefine how goods are traced, secured, and transacted [3], [4], [5].

This research paper delves into the dynamic integration of the Ethereum blockchain, exploring its pivotal role in enhancing goods security and fostering transparency within the realm of commerce. At the core of this exploration lies an innovative approach that harnesses the unique features of the Ethereum blockchain. This approach is built on two essential pillars: the establishment of secure goods storage practices and the implementation of an effective anti-theft strategy. Both pillars leverage Ethereum's cryptographic capabilities and consensus mechanisms, providing a comprehensive framework to ensure transaction authenticity and prevent theft.

By delving into the technical intricacies of this approach, this paper aims to unravel the synergy between Ethereum's blockchain architecture and the imperatives of goods security and anti-theft measures. It will explore the integration of unique identifiers, advanced encryption techniques, and the immutable nature of the Ethereum blockchain to create a system that empowers both buyers and sellers.[8] Additionally, this paper underscores the potential of Ethereum's transparent ledger to establish unprecedented levels of trust and accountability within the transaction ecosystem.

In 2021, global retail e-commerce sales reached a staggering 5.2 trillion U.S. dollars, a figure expected to grow to USD 8.10 trillion by 2024. [1], [2] In this context, our research paper introduces a novel solution aimed at enhancing transaction security. We propose the use of RSA symmetric key encryption and advanced hashing techniques to bolster security. [6], [7] This approach not only elevates the security level but also ensures data integrity. The main contributions of this paper are as follows:

- We present a new e-commerce transaction system based on blockchain technology. This system provides separate login access for sellers and users, storing data within a smart contract that is retrievable. Each user is allocated a separate storage area for their data.
- Our paper proposes the use of the InterPlanetary File System (IPFS) with Content Identifier (CID) for secure image storage.
- All data is integrated with a timestamp to generate a unique ID for each product, enhancing traceability.
- We propose a solution to identify product theft, which can be detected when a seller attempts to sell a stolen item.
- Our research addresses scenarios where a user gifts a product to their neighbours and later decides to sell it. The system identifies the product as a gifted item, ensuring transparency and accountability in transactions.

In summary, this research paper stands as a testament to the transformative potential of technology in reshaping traditional practices and fostering a new era of trust and transparency in the global marketplace, especially as the world of commerce continues to evolve rapidly.

2. Related Works

Many researchers have explored various techniques and solutions to enhance the security and efficiency of ecommerce transactions, particularly within decentralized systems. Zhou et al. proposed a blockchain-based approach to address issues related to fake ratings and secure product information storage, thereby fostering trust in online shopping.[9] Xiao et al. introduced a decentralized ecommerce system that leverages blockchain technology and Interplanetary File System (IPFS) for improved reliability, emphasizing reputation evaluation and incentives.[1]

Shamili and Muruganantham developed a decentralized web application for e-commerce, deploying Ethereum's

blockchain for secure data storage. They focused on enhancing data security and user convenience, aligning with the growing interest in blockchain adoption by major online shopping platforms.[10] Ranganthan et al. proposed an Ethereum blockchain-based application to resolve issues associated with centralized online marketplaces, including merchant blocking, listing fees, and user data privacy.[11]

Alizadeh et al. tackled the challenge of efficiently storing large amounts of data in a decentralized and secure manner using blockchain technology, combining it with IPFS distributed hash table (DHT) technology.[12,13] Li et al. introduced a data-sharing platform that merged blockchain and IPFS technology to meet the growing demands of data sharing, focusing on data security and efficient access.[14] Bhalerao et al. established global connections among hospitals, patients, and doctors using a public Ethereum blockchain. They prioritized patient privacy and secure storage, employing RSA asymmetric key cryptography and IPFS for decentralized, immutable storage. Our research uniquely addresses a crucial aspect overlooked by many existing e-commerce systems of handling the theft cases.[15]

Aggarwal and Yadav highlighted the potential of blockchain technology, including the InterPlanetary File System (IPFS), to enhance data security across various sectors, such as ecommerce, transportation, and social networking. Their focus was on secure data transmission and data integrity.[16] Shakila and Sultana introduced a decentralized application on the Ethereum blockchain, emphasizing cost-efficiency, user control, and security enhancements.[17] Izdehar et al. introduced a novel approach using Ethereum smart contracts for secure data storage and management, emphasizing user access controls and optimized gas consumption. They incorporated tag-based query systems and IPFS integration for improved efficiency.[18] Chisthi proposed a method for direct smart contract-based blockchain data access, enhancing data transparency through indexing transaction details.[19]

Kumar, Gulshan, et al. introduced the 'PRODCHAIN' system, utilizing blockchain technology to enhance the sustainability and traceability of various e-commerce products. This innovative system employs lattice-based cryptographic methods to simplify product tracking and introduces a consensus process called Proof of Accomplishment (PoA) based on ratings. Experimental studies conducted on the Ethereum network demonstrate the efficiency of PRODCHAIN, particularly in terms of latency and throughput. This research contributes valuable insights into the potential of blockchain technology to revolutionize future supply chain developments.[20]

However, despite these notable contributions, most of the proposed decentralized e-commerce systems have primarily focused on facilitating transactions between buyers and sellers. A significant gap remains in addressing cases of theft or fraudulent activity involving valuable items. In this context, our research introduces a groundbreaking approach that not only enhances user safety and overall security but also provides an efficient mechanism to detect stolen items. Unlike existing systems, our approach is tailored to efficiently handle theft cases, ensuring that if a stolen item is attempted to be sold to a seller, it can be promptly detected and addressed. This emphasis on comprehensive security and handling of theft-related scenarios marks a significant advancement in the field of e-commerce and blockchain technology.

3. Methodology

3.1 Proposed Architecture

The proposed approach is centred around leveraging the Ethereum blockchain to enhance goods security, streamline ownership verification, and establish a robust anti-theft mechanism. This novel system encompasses secure product registration, image storage using IPFS and a seamless process for product resale.

The core components of this approach, each contributing to the holistic solution, are detailed below:

- **Secure Product Registration and Ownership Transfer:**
    a) **Seller's Product Entry and Initiation:** The process begins when sellers input key details of the product, encompassing the seller's information, buyer's name, email, and comprehensive product

particulars. This initiated entry prompts the system to create a time-stamped record that includes all the provided details. Subsequently, a unique product identifier is generated by applying the `keccak256` cryptographic hash function to this time-stamped record. This hash, serving as a tamper-resistant and verifiable identifier, is then sent to the user via email for subsequent ownership verification. This secure and transparent approach ensures the integrity of the data and provides a robust means of confirming product ownership in the blockchain-based system.

b) **Buyer's Account Registration**: Upon receiving the unique identifier via email, buyers proceed to register the provided product details in their personal account within the Ethereum blockchain. Each buyer is allocated a dedicated storage space where their owned product records are maintained.

c) **Ownership Transfer and Verification:** The buyer's registered product details are cross-verified against the original seller-entered information. Upon successful matching, ownership of the product is seamlessly transferred to the buyer, securely recorded on the Ethereum blockchain.

Implementation of the proposed model is shown in below figure 1.

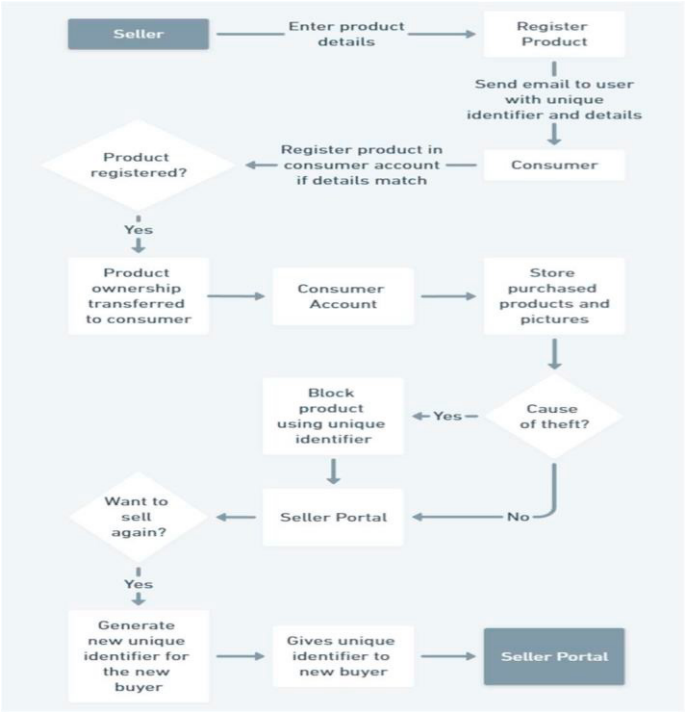


Fig.1. Implementation of the proposed model

- **Image Storage Using IPFS:**

a) **Integration of Interplanetary File System (IPFS):** To enhance the comprehensive record of purchased products, images of the products are stored using IPFS, our proposed approach introduces an innovative method for secure and efficient image storage and retrieval within the Ethereum blockchain ecosystem. Leveraging the InterPlanetary File System (IPFS) integrated with Content Identifiers (CIDs), we have devised a groundbreaking solution to optimize image handling. In this method, images are initially divided into chunks, with each chunk being assigned a unique CID, serving as a content-based address. Rather than directly storing complete images on the Ethereum blockchain, our approach focuses on storing these CIDs, which act as secure references to the image's location within the IPFS network. This approach effectively mitigates the challenges associated with on-chain image storage, significantly reducing costs and ensuring

scalability. To retrieve images, users or applications need only submit the CID to IPFS, which proficiently orchestrates the retrieval and reassembly of images from its distributed network of nodes. Our proposed method offers a decentralized, cost-effective, and secure solution for image management within Ethereum-based applications, maintaining data integrity while addressing blockchain resource consumption concerns.

b) **Decentralized Image Storage:** In our proposed approach, each image linked to a purchased product undergoes encryption and is subsequently distributed securely across the nodes within the IPFS network. This decentralized storage method enhances data security and resilience, aligning seamlessly with the core principles of decentralization and immutability that underlie blockchain-based systems.

- **Anti-Theft Mechanism and Resale Facilitation:**
  a) **Anti-Theft Measures:** In the unfortunate event of theft, the buyer possesses the capability to block the stolen product using its unique identifier. This blocked status is immediately communicated to the seller's portal, enabling prompt action, investigation, and support in recovering the stolen item. Moreover, when a thief attempts to sell a stolen item to a seller, the system will promptly identify the item as stolen, preventing any unauthorized transactions and reinforcing security measures. This comprehensive approach ensures that theft-related scenarios are efficiently addressed, contributing to a safer and more secure e-commerce environment.
  b) **Efficient Resale Process:** For buyers who intend to resell a product, the system generates a fresh unique identifier. This identifier is then shared with prospective buyers, ensuring an efficient and secure resale process.
  c) **Transparent Transaction History:** Every transaction, encompassing initial purchases and subsequent resales, is meticulously recorded within the seller's portal. This transparent transaction history empowers the seller to ascertain the product's journey, even after it changes hands multiple times.

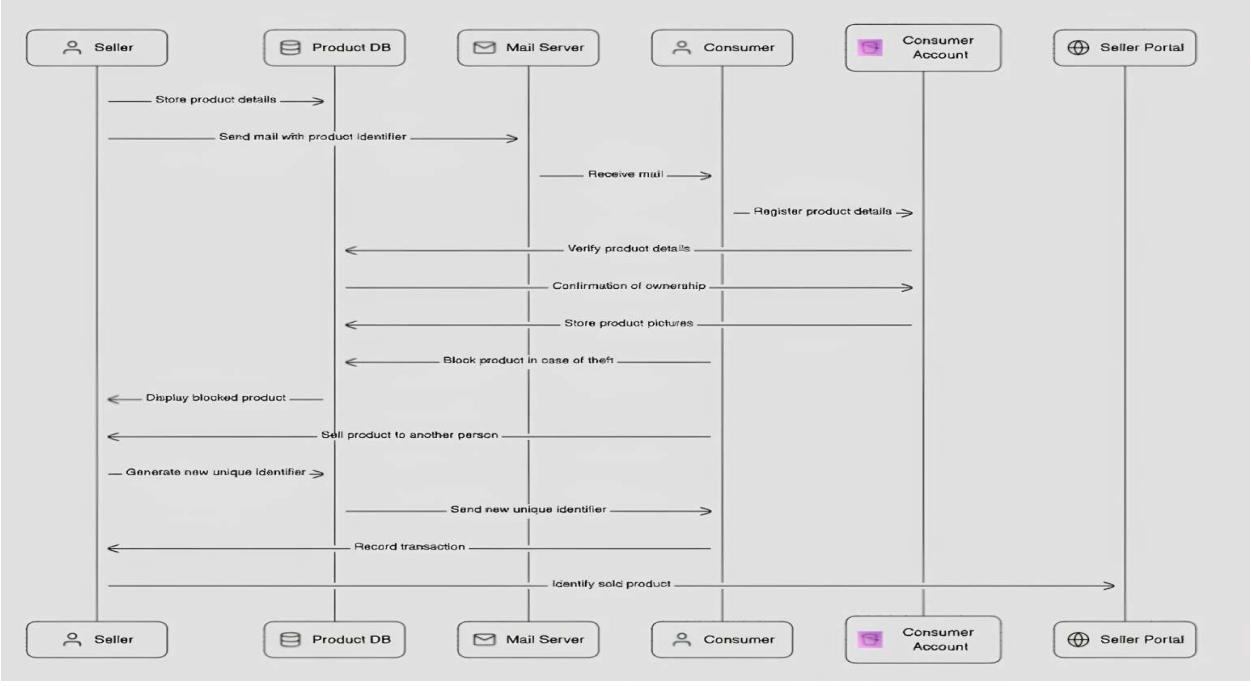


Fig.2. Flow from seller's perspective

Flow from the seller's perspective is shown in the above figure 2. The proposed approach seamlessly integrates blockchain, IPFS, and cryptographic techniques to create a comprehensive solution that redefines goods security and transparency. Through secure product registration, decentralized image storage, and a robust anti-theft mechanism, this approach enhances buyer-seller interactions

while ensuring accountability and trust throughout the goods transaction lifecycle. By harnessing the Ethereum blockchain's capabilities, this innovative approach lays the foundation for secure, tamper-proof, and transparent commerce.

3.2 Implementation

The realization of the proposed approach involves the following key steps:

- **Smart Contract Development and Deployment**: Smart contracts are created using Solidity programming language and deployed on the Ethereum blockchain. Each contract encompasses functions for product entry, ownership transfer, anti-theft actions, and resale preparation.
- **Cryptographic Security:** Seller and buyer details are encrypted using RSA asymmetric key cryptography before being stored in smart contracts. [6.7] This ensures that sensitive information remains confidential within the blockchain. Additionally, unique product identifiers are generated by applying the keccak256 cryptographic hash function to product details. This hashing process creates a tamper-resistant fingerprint of the product information, which is stored in the smart contract.[3] This combination of encryption and hashing techniques enhances data security, privacy, and the integrity of transactions within the blockchain-based system. It's important to manage the RSA key pairs securely to safeguard the confidentiality of the encrypted data and maintain the overall security of the system.
- **IPFS Image Storage:** Product images are encrypted and stored on the IPFS network. Hash addresses of images are stored within the smart contract, ensuring secure and decentralized image storage.
- **Anti-Theft and Resale:** Buyers can block stolen items by triggering corresponding functions in smart contracts. For resale, buyers initiate the process in their Ethereum wallet, generating new unique identifiers for secure transactions. 5) Seller's Portal Interface: The seller's portal interacts with Ethereum smart contracts, displaying transaction history and product details. It provides an accessible view of each product's journey.
- **User Feedback:** After a successful transaction, users have the capability to provide comments and feedback for the purchased products, which are securely stored through the utilization of smart contracts.
- **Security Audits:** Smart contracts undergo thorough security audits using specialized tools. Periodic checks ensure the integrity of data stored on IPFS, maintaining data tamper-resistance.

This implementation blueprint ensures secure and transparent goods transactions, fortified against theft incidents, and leverages the potential of blockchain technology to redefine commerce.

4. Results and Analysis

The implementation of the proposed approach yielded promising outcomes, affirming its efficacy in enhancing goods security, ownership verification, and transaction transparency. Notable results include the successful deployment of smart contracts for each product entry, showcasing the system's ability to manage transactions securely. The application of RSA asymmetric key cryptography effectively protected sensitive information, maintaining the confidentiality of seller and buyer details.

Integration with IPFS enabled decentralized and secure image storage, ensuring data integrity and accessibility. The anti-theft mechanism allowed users to promptly block stolen items using unique identifiers, enhancing overall security. The seamless resale process generated secure identifiers for secondary transactions, ensuring a smooth transition of ownership. The seller's portal efficiently interacted with smart contracts, providing sellers with a transparent view of transaction history. These results collectively demonstrate the potential of the approach to transform goods transactions, addressing security concerns and offering accountable commerce.

4.1 Testing of smart contracts

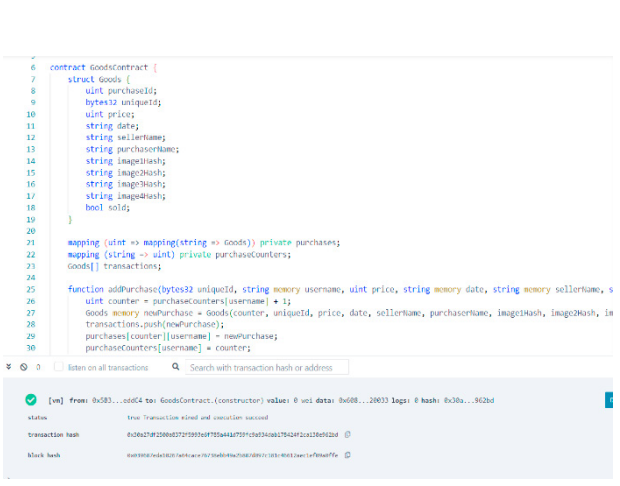
Prior to conducting testing procedures, it is imperative to compile and organize the smart contract. In this regard, the smart contract was compiled and tested within the Remix platform. Subsequently, upon successful compilation, the smart contract was promptly deployed. To emulate transaction-related activities, an account within the Ganache environment was utilized. The outcome of the compilation process is illustrated in Figure 3.

a



Fig.3(a). Smart contracts compilation tests for Seller's product entry

b



c

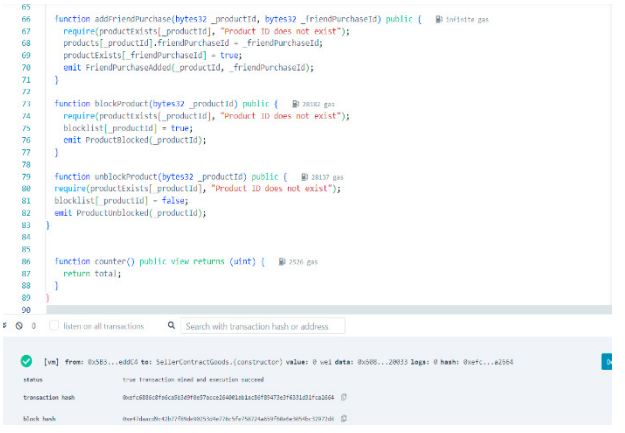


Fig.3(b). Smart contracts compilation tests for storing and retrieval of information (c). Smart contracts compilation test for blocking and unblocking the product.

After completing the transaction, the transaction information was recorded on the blockchain. As depicted in Figure 4, we can observe the decoded input and output.

a



Fig.4(a). Output for seller's product entry

.b



Fig.4(b). Output for Successful purchase

4.2 Gas Consumption Analysis

Every transaction conducted within the smart contract incurs a certain level of gas consumption, representing the computational resources utilized during these operations. Figure 5 provides a visual representation of the gas consumption associated with each transaction. This figure serves as a crucial indicator of the computational efficiency and resource demands within our smart contract-based system. Analysing these gas consumption patterns offers valuable insights into the efficiency and cost-effectiveness of our blockchain-based e-commerce solution, facilitating a deeper understanding of its real-world viability and scalability.

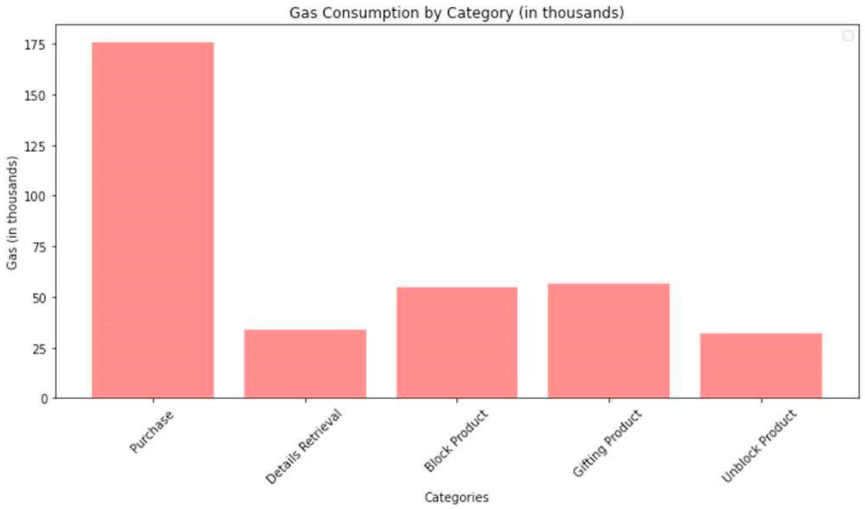


Fig.5. Gas consumption in smart contracts

5. Discussions

In the realm of E-commerce research and development, the current lag in addressing product theft represents a noteworthy gap in the anti-theft approach. While advancements in secure data storage and cryptographic techniques have significantly improved the overall security and integrity of online transactions, the specific issue of product theft remains an underexplored challenge. The innovative work of researchers like Shamili, Muruganantham, Aggarwal, and Yadav, who emphasize enhanced data security and secure data transmission through blockchain technologies, offers promising avenues to bridge this gap and create more comprehensive anti-theft solutions for valuable items, such as ornaments, in the E-commerce ecosystem. Addressing this lag is crucial to ensuring the continued evolution and enhancement of security measures within the E-commerce landscape.

Our methodology introduces a novel perspective in this domain by addressing the significant challenge of product theft within the context of E-commerce transactions. In contrast to previous approaches that primarily concentrate on the technical aspects of secure data storage and retrieval, our method emphasizes the real-world implications of theft in online commerce. We propose a mechanism where users can proactively block their stolen products using unique identifiers associated with each item. This innovation not only enhances the security of the blockchain-based Ecommerce ecosystem but also empowers users to take immediate action in the unfortunate event of theft.

Moreover, our methodology introduces a crucial layer of transparency and accountability by rendering blocked items visible to all seller portals. This feature ensures that even if a stolen product is brought to a seller for resale, it can be promptly identified as stolen. This not only protects potential buyers from unwittingly purchasing stolen goods but also aids law enforcement agencies in tracking and recovering stolen items. It represents a significant leap in enhancing the overall trustworthiness of online transactions.

Furthermore, our approach extends beyond theft prevention. We also account for scenarios where users may wish to gift their products or undertake other legitimate transactions. All of these activities are meticulously recorded in the seller portal, creating an immutable and transparent transaction history. This not only benefits users by providing a comprehensive overview of their product's journey but also fosters trust within the blockchain-based E-commerce community. In essence, our methodology not only bolsters security but also enhances user experiences, making it a valuable contribution to the ongoing discourse in Ecommerce blockchain technology.

6. Conclusion

The journey of implementing a goods storage and anti-theft approach on the Ethereum blockchain has uncovered significant insights and possibilities for revolutionizing commerce security and transparency. This research paper embarked on a quest to enhance goods security, streamline ownership verification, and elevate transaction transparency through the amalgamation of blockchain, cryptography, and decentralized storage. The successful implementation of smart contracts, coupled with cryptographic security measures, has demonstrated the capacity of the approach to create a secure and tamper-resistant ecosystem for goods transactions. The integration with IPFS has showcased the potential of decentralized storage in enhancing data availability and integrity. The innovative anti-theft mechanism empowers users with swift action against theft, reinforcing goods security. The seamless resale process efficiently transitions ownership, contributing to a comprehensive commerce lifecycle. The seller's portal facilitates transparent transactions, fostering user trust and informed decision-making.

However, this journey also illuminated the road ahead. Challenges such as scalability and gas costs within the Ethereum network demand continued exploration. Enhanced security measures and refined user experiences are imperative for sustaining system integrity. Collaborative efforts to navigate regulatory landscapes will be instrumental in realizing real-world adoption.

In conclusion, this research paper sets the stage for a transformative paradigm in goods transactions. The approach's potential to enhance security, transparency, and accountability is undeniable. As we continue to refine and expand the approach, we embrace the promise of a future where commerce thrives on the pillars of trust, efficiency, and innovation. Looking ahead, the future scope of this approach involves addressing scalability challenges, optimizing gas costs, further enhancing security protocols, and fostering collaboration within the regulatory framework. Through ongoing research, collaborative efforts, and technological evolution, the vision of secure and transparent goods transactions is not just a possibility but a compelling reality on the horizon.

References

[1] Yuanyuan Xiao , Chuangming Zhou , Xinpeg Guo , Yafei Song and Chen chen “A Novel Decentralized E-Commerce Transaction System Based on Blockchain” Applied Sciences(12):5770

[2] Sk prashanth, S China Ramu, K. Raghu Ram Mohan Reddy, Sukanya ledalla, Raman Dugyala, G. Vijendar Reddy, and Yerragudipadu Subbarayudu (2023) “DDSBM: Distributed Decentralized Smart Contracts On Block Chain Marketplace For Books Through Ethereum” European Chemical Bulletin (12)

[3] Gousia Habib, Sparsh Sharma, Sara Ibrahim, Imtiaz Ahmad, Shaima Qureshi and Malik Ishfaq “Blockchain Technology: Benefits, Challenges, Applications, and Integration of Blockchain Technology with Cloud Computing,” Future Internet (14):341

[4] Md Ashraf Uddin, Andrew Stranieri, Iqbal Gondal, and Venki Balasubramanian. “A survey on the adoption of blockchain in IoT: challenges and solutions” Blockchain:Research and Applications(2):100006

[5] Horst Treiblmaier and Christian Sillaber (2021) “The impact of blockchain on e-commerce: A framework for salient research topics,” Electronic Commerce Research and Applications (48):101054.

[6] J.S.Coron “What is cryptography?” IEEE Security & Privacy Magazine (4):70-73

[7] Sandeep Kumar Panda and Suresh Chandra Satapathy, “An Investigation into Smart Contract Deployment on Ethereum Platform Using Web3.js and Solidity Using Blockchain,” (1407):549-561.

[8] Frank Hofmann, Simone Wurster, Eyal Ron and Moritz BohmeckeSchwafert “The immutability concept of blockchains and benefits of early standardization ,” IEEE Xplore,1-8.

[9] Zhili Zhou, Meimin Wang, Ching-Nung Yang, Zhangjie Fu, Xingming Sun and Janathan Wu “Blockchain-based decentralized reputation system in E-commerce environment,” Future Generation Computer Systems(124):155-167

[10] Shamili, P., and B. Muruganantham.(2022) “Enhancing the Decentralized Application (Dapp) for E-Commerce by Using the Ethereum Blockchain.” Proceedings of International Conference on Recent Trends in Computing,679–694.

[11] Ranganthan, Ram Dantu, Aditya Paul, Paula Mears and Kirill Morozov (2018) “A Decentralized Marketplace Application on the Ethereum Blockchain.” 2018 IEEE 4th International Conference on Collaboration and Internet Computing (CIC),

[12] Alizadeh M, Andersson K, and Schelén O. (2020) “Efficient decentralized data storage based on public blockchain and IPFS,” In 2020 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE),1-8

[13] S. Jianjun, L. Ming and M. Jingang, (2020) "Research and application of data sharing platform integrating Ethereum and IPFs Technology," 19th International Symposium on Distributed Computing and Applications for Business Engineering and Science (DCABES),279-282.

[14] Weijing Li, Zicheng Zhou, Wen Fan, and Juan Gao (2022) “Design of Data Sharing Platform Based on Blockchain and IPFS Technology.” Wireless Communications and Mobile Computing (2022),1–7.

[15] R. Bhalerao, P. Gite, P. Patil, R. Gupta and S. Singh, (2023) "Decentralized Ehealth Patient Record Management System using Blockchain and IPFS," 2023 International Conference on Innovative Data Communication Technologies and Application (ICIDCA),781-785.

[16] Aggarwal, Kapil, and Santosh Kumar Yadav. (2021) “Decentralized Cloud and File System for Blockchain Environment.” Springer EBooks,331– 339.

[17] U. K. Shakila and S. Sultana, (2021) "A Decentralized Marketplace Application based on Ethereum Smart Contract," 2021 24th International Conference on Computer and Information Technology (ICCIT),1-5.

[18] W. Zhao, Izdehar M. Aldyaflah, P. Gangwani, S. Joshi, H. Upadhyay and L. Lagos, "A Blockchain-Facilitated Secure Sensing Data Processing and Logging System," IEEE Access (11):21712-21728.

[19] M. S. Chishti, F. Sufyan and A. Banerjee, "Decentralized On-Chain Data Access via Smart Contracts in Ethereum Blockchain," IEEE Transactions on Network and Service Management (19):174-187.

[20] Gulshan Kumar, Rahul Saha, William J. Buchanan, G. Geetha, Reji Thomas, Mritunjay Kumar Rai, Tai-Hoon Kim, and Mamoun Alazab, (2020) “Decentralized accessibility of e-commerce products through blockchain technology,” Sustainable Cities and Society (62):102361.